

ActInf Livestream #015.2 ~ “Free-Energy Principle, Computationalism and Realism: a Tragedy”

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hello everyone welcome to the active inference live stream from the active inference lab it is active inference live stream 15.2 today and it is february 9th 2021 welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference you can find us on our website twitter email youtube discord or keybase this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we will follow good video etiquette for live streams today in the active inference live stream you can check out the calendar for 2021 at this link on the left side and we're here with the red arrow on february 9th for 15.2 this is our second group discussion with the two authors of an awesome paper about realism and computationalism and instrumentalism and so this follow-up discussion we're going to be probably taking it in a lot of different directions bringing up some ideas that we heard about in earlier 15 discussions and sending us off with momentum into the future of 16 and beyond in 15.2 we're going to go around and hear from everybody in the introduction section and the warm-up and then we will jump around go to some questions that we had from last week and anything else that people are thinking about will be awesome to hear so for the introduction and warm up i'm daniel i'm a postdoctoral researcher in california and i will pass it first to shannon hey i'm shannon i'm a phd student at the university of california in merced and i'll pass it to stephen hello i'm stephen i'm here in toronto i do a practice based phd and i will pass it over to dave okay i'm a retired i.t guy with a background in process philosophy and uh natural language processing living in the philippines and let's go to ines hi everyone so my name is inish i am based in the berlin school of mind and brain and i work in philosophy of cognition and i'm very happy to be here thanks daniel pass it to blue everyone i'm blue knight i am an independent research consultant based out of new mexico and i will pass it to alex thanks hi everyone i'm alex i'm in moscow russia and i'm a researcher in systems management school and i pass it to scott hi folks i'm scott david i'm the

director of the information risk research initiative at the university of washington applied physics lab i've been a lawyer for 30 years and we engineer legal and socio-technical systems together thanks is there anyone else to pass to daniel thomas hi everyone um my name is thomas van s and i work in antwerp on a philosophy of cognition cool well this will be a really fun discussion so let's just get warmed up with kind of the pivot point with the phase transition from all of our disparate lives whatever we were thinking about before this conversation and get warmed up into the paper and so the three warm-up questions are on here anyone who raises their hand can just take a stab and it'd be cool to hear from everyone what is something you're excited about today what is something you liked or remembered about the paper and then what is something that you're wondering about or would like to have resolved by the end of today's discussion so while people are raising their hand one thing i was curious about is what leads one to be an instrumentalist or a realist like how do we um make that choice what kind of sends us down one road or another or is it something that switched back and forth just something i was curious about anyone else have a um thought on one of these warm-up questions or just something that stayed with them from the paper um dave and then anyone else who raises their hands yeah i would hope that folks don't get stuck in having to choose between instrumentalism and really getting into the meat of causation in any particular topic that you're interested in uh you gather data by looking at the envelope of a phenomenon and when you've gotten enough data you just say well what is it like to be one of them and uh one hopes you've got enough mirroring or modeling of the relations that you can start coming up with some [Music] some mirrored explanations of how to what is it like to be uh a thermostat cool thanks anyone else steven yeah um yeah i think that this question around thinking about using an instrument is quite interesting sort of that embodied sense of like i'm probing something with an instrument um as opposed to maybe using a tool which maybe is a bit more like i can hold that in my hand you know i know what my tool does but my instrument there's like a little bit of a gap so i i like that way of thinking about how we are working with something between us and the world and where does the idea of a system go in there or where does the idea of a tool or where does the idea of you know it actually in some ways is kind of realist in the sense of how we try and think about things physically so i quite like that ironically shannon and then anyone else yeah so i don't know what it's like to be a thermostat um but that made me think of the dynamics of a crowd that consists of a lot of individual people and no one person in the crowd know what knows what it is like to be the whole crowd and you can identify certain behavior that's happening at the level of the crowd and we can be

super instrumentalists and say this crowd is navigating a space or something um and that sits fine with me but somehow when i transition to neurons in a brain i want to say that there is something realist and we don't have to be instrumentalist about the brain we can say this population of neurons is representing some activity but i don't want to say that about a crowd but if you put those two side by side i should i should have the same view against both of them interesting thanks yeah we line up all these kind of mappings and then it's almost like um well the linear regression is an instrumental um way of doing statistics so then if we're going to be serious about instrumentalism could we replace linear regression with some other statistical technique or does it come down to the details of the instrument in s and then anyone else who raises their hand yeah i was thinking about um dave's remark actually um which i i also agree with and i think that dave's remark allows us to say a little bit more about why we need to be careful about being realist and uh that is um with the fact that if we're going to model say a cognitive phenomenon say perceptual learning on something or a phenomenon that is cognitive that entails also what it is like then probably in modeling that cognitive phenomena and we are going to have to take a very realist stance because you won't be able to capture the what it is like so in a way if we are taking the realist path and saying that well my model is not only modeling the cognitive phenomenon but actually that's how the phenomenon works by using these models and these representations then we are we also are committed to the implication that um that particular phenomenon does not have or is not permeated by any phenomenological aspects because we are applying a reductionist model and taking a realist perspective so taking a release perspective is not going to be useful if we want to account for the phenomena the phenomenology aspect of the cognitive phenomenon yeah thanks for that um and anyone else can raise their hand i have a question for the authors so the title talks about computationalism but one of the main tensions in the paper is about instrumentalism and realism but the title has computationalism and realism so what is the relationship between yeah ns yeah so um i can say something about that um so in computationalism we can take either instrumentalism or realism which is cool it's fine um one way to think about this is that i can be uh completely in agreement with computational science and use all the apparatus and machinery that computation science has to offer at our disposal and i can use that from an instrumentalist perspective now the other way is being realist within computationalism and i'm going to say that it's not only a tool that i use this computational machinery but it's also um the way that in this case the target system which is the brain or cognitive activity works so these are the two options of um or the two uh um options that we have within the computation

within the the typical um the typical standpoint is to be a realist about it that's the history of philosophy of mind is to apply a realist attitude towards our scientific models of the brain or cognitive activity so it's not only that i have a representation or i'm developing a representation of what i think the brain is doing in generating uh these data that i have access to but i also think that this representation is also in the brain that's also how the brain works by having these sort of representations permeated with all these cultural constructs that we have like numbers logics uh bayesian inference concepts and all of that is in the brain too cool thanks thomas and then anyone else yeah um so i think that's that's exactly right and um last time it came up as well that uh it seems like only philosophers are worrying about this or making this uh distinction or anything um but the thing is that you see that in in certain scientific projects as well um when people uh i've heard anyway of a research project where people are trying to identify particular priors in the brain are trying to find neural correlates of the priors in bayesian probability distributions um that only makes sense like the entire research project only makes sense if you think the brain is literally engaging in bayesian inferences so it really matters in that sense also for many scientific projects whether you take an instrumentalist or realist attitude thanks yeah i think it's something we can kind of tease out and return to is what are the implications for where we fall out on this instrumentalism and realism continuum or what are the consequences for different projects or for different kinds of explanations predictions models um is it possible to use bayesian statistics instrumentally on the brain and be totally neutral and talk about it like it's a uh like uh you can model it usefully as a computer or as doing bayesian statistics but then not take the realist turn and say that it is doing that or do those two tracks run so close to each other that maybe it's a just a slip from one to the other and what would be the consequences of that so stephen and then anyone else i'd be curious to know whether you've got the idea that when you are trying to probe the brain or try to understand it then you take an instrumentalist perspective i was wondering if there is like a level at which realism is sort of still seen to be there like even if it's in a sense that can't be seen i.e the free energy principle because it's a principle and it's and the idea of non-equilibrium steady states or the fact that it's you know that you need the dynamics to be happening rather than it being a system that can be just snapshotted is that in principle form still coherent to have as like a relatively realist interpretation it's just that doesn't necessarily mean that any of your models are realist they're just instruments or is it not that is there is there something like that kind of interaction at a more foundational level cool question thank you um dave and then anyone else yeah on that point i would recommend you look at the um

mark some lecture series from 2013 on the conscious ego he is both a neuroscientist and a psychoanalyst he is also currently collaborating with carl fristen and is in like two weeks bringing out a book that uh integrates dream study psychotherapy and um bayesianism solm s cool any other thoughts on the introduction or we can um just start with some of questions that we had raised as a group last week so yeah sounds good we'll just kind of i guess walk through a couple of these questions people can take it in any direction though so there's not too much of a linear ordering here but we'll just walk through them so this first question um several people spoke to it last week and it was related to conscious awareness and to self modeling so i um was reminded of this with steven's question about instrument and tool so like do we have an i is there something that isn't you know is the uh maybe i'll pick a different word that's not you know literally the same e y e or i but let's just say like um for tactile sensation is it that we have proprio sensation or that we are proprio sensation or do we make a model of proprio sensation or are we agnostic on how it really is with proprio sensation but we're just gonna model it a certain way there's a lot of directions that we can take this but i think that the interesting question that was raised is where does the self awareness come from and then is a self uh enacting these types of uh inference schemes or is it as if they're in acting schemes how do we think about that for ourselves how do we think about that for other systems um yep ines then chan um i think that um one point uh at which we can start this is by asking ourselves um what is modeling what are the motivations for us to uh get around and uh construct a model of something so i would say um that we model things we do not have a complete access to we are somehow missing something about you have an object of study but somehow we are missing um some part of it that is puzzling to us and we don't have a a direct access to it almost like saying we don't have the true posterior so we need to find it by um uh coming up with theories um and uh by thinking about uh what would be uh what would make sense um to explain that particular phenomenon that we are puzzled about so i think this is important for this question of self-modeling because then it raises the question of whether i have access to the self or not because if we think that however when we answer this question is going to be very relevant to however we think this question if we think that we the self is something that is hidden away and we don't have a complete access to it then we could perhaps think about that oh there's motivation to model the self how is a different question if we think that no i have a direct access if to anything to the self on a conscious level um i might not understand everything about myself or the self um but i do have a direct access in which case perhaps there is no motivation to model it um so yeah i just think this is a nice way to to start thinking about this

thanks shannon and then thomas i'm actually going to hold my comment until a little bit later all right cool thomas yeah um so i think two other very important questions here i don't know maybe there's another one what is access um what do we mean with the conscious level what is the self that we are supposed to be modeling or not modeling i think that that might be it and and this seems a little nitpicky and typically philosophical i can imagine but it really is very important that we get clear on these things because some people will have a very very narrow definition of what you have access to um which will very quickly kind of mean you will need to represent or model just about anything um whereas if you have a broader definition of what we have access to that's going to change the story completely um the same thing goes with the self some people have different varieties of the self there's this sort of pre-reflexive self that is just sort of self-hood some people call it that you just have a feeling of being yourself and as an acting person and there's a narrative self that's sort of the story you make up about yourself like what sort of person am i i'm etc those are all going to be enormously important questions to answer before we can really talk about what self-modeling on the even is i think great point it is very philosophical to to take one question and just fractalize ask sub questions but it's so true it really does matter if we think that we're going to explain some consciousness in one definition then we might be off base so i guess since these are big questions what might the free energy principle or active inference do to help us resolve or at least approach these ideas shannon and then it could be on that or or anything else than anyone else yeah so coming up from what you said do we have proprioception like do we have a certain sensation or are we that sensation are we proprioception um and anest's question like what are the motivations for modeling um and we model if we're somehow missing the full picture um and this brings me to like in yakipovi's book predictive mind right the brain doesn't have access to the sensory environment the brain has access to electrical firing and so it's trying to model what in the environments could have caused or what um what could have caused that certain pattern of electrical firing and back to last week when we were talking about representations having some sort of truth conditions if we do you know electrically stimulate um part of your somatosensory cortex and then you your conscious self model is that i feel a sensation of somebody touching my hand because you stimulated that part of my somatosensory cortex so on the like conscious self-model level i'm making sense of what my neurons are telling me and my neurons if we're asking what is it like to be a neuron um is making sense of the electrical signals that it's receiving by saying they must have come from this area of the body because usually when these bonds are stimulated in that way that corresponds to a sensation from the

sensory environment from this part of the body and if we think of it that way then maybe we could say that set of neurons is predicting or doing some sort of modeling of the world because it's using these signals as a stand-in for the world these electrical signals as a stand-in for the part of the body that's usually associated with that or we could be instrumentalists about it and just say that we're using our math to describe how the neurons are doing interesting and great to introduce that experimental result that when you do you know give a little electricity the person will identify that as being experienced a certain way so um in s and then stephen um okay so i just wanted to address a few of um of thomas's questions and just to give a little bit of context with these uh particular questions coming from i think it was from mark's uh discussion with the discussion remark last week and we were talking about uh the body uh being modeling itself and then this came up what about self-modeling uh on the what would that be um and here i think that there are two things um so uh thomas asked what is more i think i asked okay doesn't matter so um modeling um here is important because how is it that the self would be modeled is important because all of us understand what is modeling from a scientific perspective we develop our tools and we know how things are modeled right depending on the target system i'm going to use certain set of tools uh that's going to be dictated by the system that i'm targeting okay now we know what that kind of modeling is i don't know what modeling is when we turn to the self i have no idea what that looks like is it asian inference what kind of tools are we talking about that um the nervous system or the body or even myself and this is the point of this question or even myself on a conscious level and by conscious level i don't even like the expression conscious level um here i was more thinking of the agent level so the agent that interacts with the world and has um representations and things about the world is capable of drawing mapping things that's um that's a sort of a much more sophisticated kind of interaction with the world that entails language concepts logics uh skills and that's the agent level right so this is like lifting up this into the agent level and thinking about okay how is it that self model modeling would be motivated would it be motivated at all um and if it is how is it that it happens because um on an agent level and i think this is this is a a a point that might elucidate things for us on an agent level is an agent ever engaging with beijing inference i haven't learned bayesian inference until very late in life so before i learned beijing inference i wasn't able to model myself i guess so that's the point so even on an agent level for me to interact with the world by using representations because i can on an agent level i don't think anyone has an issue with that not even radical in activism not even any other an activist i don't think um so on a conscious level on an agent level where we can actually

engage in representations and use logics and use our concepts are we uh using uh these kind of uh modeling that we know what it is in science so that's one point about modeling the other one is um i think was shannon that mentioned this very nice example um and i think it's useful for us to look at uh which is um shannon said i feel a sensation of someone touching right touching touching my heart my arm or something right um so in that particular case this is the agent level that's why i think this is a nice example in that particular case i feel someone touching my arm do i go around and uh start wondering about whether i felt that touch or not because that would be starting off an inference kind of kind of thinking which we can totally do by using logics and deduction and induction and abduction all of that we can totally do that uh because we are enculturated beings now do i wonder about whether someone did really touch my arm whether do i really feel pain because if i do wonder then i engage in inference kind of thinking but that also does not come for free that i'm using beijing inference unless i have the training and i'm applying that for that purpose right so uh what i think is problematic about that is that once we start thinking about um whether someone really touched my arm in kind of like a wandering kind of way it's because we are assuming that we don't have a direct access to that particular experience somehow that particular experience is hidden from us so we need to infer whether it is really the case that i'm in pain for example right so that puts us in another yet level of experience which is like the position of a spectator the spectator does not have direct access to whatever is going on so it needs to infer what is the most um uh uh the the best explanation for that particular case yeah i just wanted to add that interesting thank you stephen and then yeah that that idea of how much you have have access to and that may relate to different types of engagement but the the the way do we have access to our multi-sensory integration so we have a sense that we are i'm here now i'm i feel myself sitting i can see i can hear i'm aware of stuff and then i'm waiting that's in different ways just in my awareness i suppose and then there's the question of like what i actually choose to do um which made me more deductive maybe more based on cues so it is interesting once you get to this conscious level is how much does the conscious level give us a way to not have to follow um ergodic kind of flows and we can kind of break out from that but when we get down to lower levels it kind of needs to be more flow it needs to basically be more biological so to speak um i don't know if that makes sense but thanks steven blue so i just wonder is the model of the self maybe the ego like are those two things the same and like does the self model necessarily need to be um like rooted in narratives uh and i think we that we kind of touched on this a little bit um last week i think scott had brought that up and then just to reflect

really on what shannon said and what he ness uh just was talking about um you know in terms of like a pain experience right like this is kind of in like the the eastern traditions pain is a very like objective experience like yes i'm feeling pain like i just burned my hand on a hot pot of water or whatever so like pain is like an objective experience but the suffering that's associated with that pain is a subjective experience and that's that's part of like your ego or identity like oh my god i feel so horrible for myself or like suffering in relation to the pain that you're experiencing is optional right it's not uh it's not something that you have to necessarily like experience and so i wonder then is suffering like inference is is that part that's the inference part or or i don't know just a thought yeah it's like the observation it's raining and if you are i'm the kind of person who's sad on a rainy day or i'm the kind of person who's happy on a rainy day it's quite disjoint from it raining or not but then sometimes when it's biological and it's about our body it's easy to conflate those two like well pain is bad and it makes me suffer because it's painful but actually there is a little bit of a step in between shannon yeah thanks for that um so yeah i wonder if there it makes sense um to say to a philosopher that we have direct access something that something emergent um but our brain is a spectator in this process so i have access to knowing that someone touched my hands i might question it if i see nobody's touching my hand but someone stimulated my brain in a certain way and i still feel that sensation i still have direct access i feel like to that sensation i don't feel like it's vertical with the world anymore or vertical with um touching on my actual hand um and then maybe on the level of of the person like blue said i can associate this feeling with pain or if this is an instant where i enjoy this feeling then i don't associate it with um the suffering i get you said of pain but my brain doesn't know that like so my neurons are still being a spectator on these sensations and are still inferring something about their causes or something about what the next sensation the next pattern of activity will be given the pattern of activity they're feeling now so on our emergent on our side of these emergent processes we can have something like direct access but within all of the individual parts that make up that emergent process we still have a bunch of spectators a bunch of neurons cool um thomas stephen and s yeah so that's that's um that's very interesting um let's see i i initially had a just a comment on blue but now also on shannon so um i think the first thing to try and understand is um what it really means to say that the brain knows anything at all um and that's actually something in holi's work that i would also quite strongly criticize because there is a weird way in which a single organ in a body suddenly becomes a an autonomous agent where it is guessing things where it is predicting things it's knowing things and doing all sorts of things that we would usually

relegate to the personal level or the agent level um this is i think fairly well known among neuroscientists the myriological fallacy ascribing agent-level behaviors to a part of the um the thing is then that the brain doesn't have access and needs to predict or because it doesn't know doesn't really make any sense anymore when you just see the brain as an organ in an organism and the organism as a whole has access um so that's that's one point the moment in what you're doing here basically you're shifting from the uh from a neurocentric way of doing cognitive science and definition of the mind to a more embodied one um and then if you're talking about the difference between pain and suffering that comes with it as blue was doing um under the inactive approach the um difference sort of falls away um and both of them are the very specific ways that the sensory stimuli and your own actions interrelate those are called sensory motor contingencies um so if you get a pin prick in your finger the pain you're experiencing is the very specific that the sensory stimuli in your finger change as you poke the needle in or the needle gets poked in if it's not you doing it or it's about squeezing a sponge and the softness of the sponge that's a very nice example um you only really feel the softness of the sponge as you're squeezing the sponge there is no softness of the sponge separate from your squeezing if you're not squeezing the sponge you won't know if it's soft and you can't feel softness so the softness you're experiencing just is the particular way that your sensory stimuli in your hand interrelates with your own squeezing so there's nothing over and above that interactivity or underneath that interactivity that makes up the experience interesting thanks steven in ass blue thanks thomas that's really well put i mean i think that's where the idea of the spectator that in itself is a you can fall into that trap of it being apart because like a spectator is assuming like i see the story coming at me and therefore the brain is having all this information come in whereas if it's part of the the mind is part of the body and the proprioception and and the dynamic with whatever is being encountered then that spectator role doesn't really make as much sense so and i think that that also then comes in with the active inference idea that the signal isn't coming in with the knowledge it's it's being created somehow well through active inference through this dynamical process so i think that's really interesting and one thing that's quite interesting to say with pain um is is pain you can only create pain you can only talk about pain through metaphor it's a stabbing pain it's an ache it's a it's um it's it's kind of a burning feeling so all pain is expressed or has to be expressed through some sort of metaphor so so um we have to utilize not just our proprioception but we're um we need to come back full loop with our sensory apparatus to create the thoughts and this is where the mental space psychology work could be quite interesting to to mix with that because that idea there is that

there's an organizing principle in the space around us that we're effectively designing a mental space around us which then becomes a part of that interface so anyway that that's uh definitely an important point thanks ernest um okay i wanted to uh just go back to um brain level and then i want to go back to the self modeling um just to make a point about that um so on the on the case of the neural level um i think that we're talking about the spectator and we're talking we're trying to pursue this path where we were motivating why the brain is a spectator and then thomas made up this nice comment um coming from um the reasons that jakob hui provides um to that direction and i just want to address that from um the free energy principle in the uh the active inference um um trend um so i think that even if we would accept that um we were under the assumption it would accept the assumption that the brain is a spectator and we were sort of agreeing here with the yakov huey um that the brain is encapsulated in the skull therefore it needs to infer whatever is going on outside we have to understand that the brain has not uh gone to school to learn asian inference or dynamical modeling so that's important and we also should not forget because this is what brings us all here we're talking about free energy principle active inference and those things so this is not something that active inference or the free energy principle will endorse right that the brain is encapsulated um uh according to the free energy principle or active inference um the brain is embedded in a multi-scale system that many people call can be captured by uh by nested markov blankets so this multi-scale system itself organizes at every single level and in function of immediate influences that we can explain for example by by by applying uh active inference and and uh bayesian uh and the leaf and that kind of thing that we all know so this means uh one important thing which is that every part of the brain is interacting okay and where there is a interaction there is no there are these causal influences and this is the principles of um effective connectivity for example which we talked about uh about last last week so i just wanted to put this out there that this sort of like spectator encapsulation is something that would not be endorsed um and i think that this is the right way to think i'm just not sure if we should take the realist um view about it the other point that i wanted to make just going back to the self-modeling and then probably just drop it but i think that this is important for us think and it wasn't mentioned um the self-modeling it's important to ask who is doing the modeling in self-modeling who is doing the modeling is it a homunculus so i think this is important because this there have been a few arguments against this idea of like of and um it's important to know always because in science we know who's doing the modeling is the scientist in self-modeling i'm not sure who's doing the modeling yeah and yeah these are my points thanks a

lot blue and then i'll ask a question from the live chat so um i don't know the the brain as a is is an interesting uh point and also like who is doing the modeling like these two maybe might go together but maybe are encompassed by a bigger question um like what does the brain know versus like what does the mind know and are these two things interchangeable like and what does the free energy principle apply to does it apply only to the brain or can we scale it to maybe include the mind right or maybe the brain is part of the mind if you take a more embodied approach or what is the mind has also been like positive to be maybe like the bioelectric field that's been out there and some work so so what is the brain what is the mind can it is the brain part of the mind i would say a pretty large part if you're looking even at the bioelectric field it's it's a big part right like so the brain has most bioelectric activity so maybe it scales to that um so so and maybe like the integration of all of the systems happens outside of the brain i mean maybe a component happens inside the brain but maybe then it scales up to include like embodied knowledge such as you know the gut brain axis or other um areas so i just was wondering really what the free energy principle as developed by first and i guess is that a brain specific or do you what do you guys think can it scale to include the mind or or are these the same thing i don't know just curious yep um i'm gonna just wait for the live chat because it looks like a lot of people have thoughts on blue's question stephen and s thomas and then i'll ask a i mean this is also where the free energy principle and the active inference side comes in so i suppose there's the active inference would be like you could say it has to be inactive so it's the whole mind-body environment dynamic but then the question with the brain is is there a level of the brain they should say it's all using free energy principle but maybe there's a level of the brain which is um you know using free energy principle approaches to create the consciousness as this kind of epi phenomenon that's sitting on top of the whole organism which is more broadly used in active inference with this with the with the um senses my feeling is everything has to go back to the mind-body environment at some level but there may be some i don't know some like slightly higher level encapsulated piece that gives you the ability to have this reflective capacity of consciousness at like the human level but i'm not sure thanks steven um ines and thomas um yeah okay so i just wanted to um address the mind brain um whether the energy principle applies only to mind or brain or both or only that um i just wanted to bring to the um the the the fact that the free energy uh is a principle that is supposed to apply to every single system that is in non-equilibrium steady states and this is the majority of the world it's very hard for us to find a system a natural system that is not um that is in equilibrium um so that's what the free energy applies to another way to say it is that free

energy applies to anything that has a markov blanket in the sense that is coupled with the environment exchanging uh energy with the environment and interacting with the environment in that particular sense obviously these things become much more complex to explain for us when uh we come to adaptive systems because levels of complexity uh sort of explode because of the the the higher levels of degrees of freedom so just to put it out there it applies to um every single system that is in non-equilibrium which is the majority of the things and then we can pursue a sort of a line of research that is about the mind brain identity where we ask questions of whether it applies only to the mind also applies to the brain or applies to both of them or tries to unify both of them in a single explanation right um so i think that um the machinery of the free energy principle with the corollary after active inference um is is very nice to to to help us understand both the mind and the brain and here by mind i'm thinking about the agent level where we we can think about decision making beliefs uh what is motivating my actions in terms of more intellectual aspects of ways that i can interact with the environment like having thoughts and beliefs uh that would be the mind and that would be the nice way where i think that the free energy principle with active inference can be very helpful um to develop um generative models for us to understand the specific psychological phenomenon that is on the agent level and then we can also apply this uh this machinery these generative models to uh the brain level which would be the neural level where we will look at the connectivity in brain activity and obviously we can explain these things by saying that the activity in the brain can be described as if there was belief update as we know uh uh with uh with active inference what i would then be a little bit more careful about is to say that um this is precisely our neuronal activity or even the agent level uh works uh when doing its own uh reasoning um to act in the environment um so that's what i would be a little bit more careful about is then the second the second step and i don't think that we lose a lot um i don't think that uh we lose the explanatory power as long as we understand that this is a tool thanks a lot thomas yeah i was um basically going to um say sort of the same thing as ines um i mean that makes it a little easier for me um i think just mostly what is also what i would like to add is the free energy principle isn't going to tell you where the mind starts and where it ends um since it does apply to just about anything that exists um you can just take any system draw your own boundaries and start analyzing and things will start popping up um when you want to think about where is the mind is it the body is it the body in an environment is it just the brain um those are philosophical questions that will need philosophical answers you can answer that in in different ways and if you want to say it's just the brain you're going to end up in sort of

always camp the predictive mind you want to say it's the the body in its environment um you're going to end up more in an inactive camp but those are separate questions that i i don't think you're going to answer just by looking at the formalism cool it's like should the title of your book be the predictive mind the predictive brain the predictive brain body mind the predictive niche you could keep going and the point is when we're taking this instrumentalist multi-scale nested markov blanket approach you just could kind of pick your title depending on what you want to emphasize so the from the chat which i think was related to this question about um who is doing the self-modeling and who am i which sounded a lot like some meditation prompts the question from cv in the chat was can spontaneously arising thought give us insight in to answer the questions in this discussion that cannot be given by any other research or other thought types if yes where and how thanks a lot so as selfs if we're thinking about this self modeling at the conscious level thomas you just said maybe there's things the formalism isn't going to resolve for us there's uncertainties we have that aren't going to be equations resolving it for us um might spontaneously arising thought or other types of reflection maybe shine some light into any of these questions so i'll just put that out there and go to just leave it up while we go to another um slide and then stephen just just to put this up because this was some related questions about the body's expectations and about where these priors or these expectations come from so any thoughts on the spontaneously arising thought or any other topic stephen and then any other hands i think this is quite interesting from the point of view of how neurophenomenology is generally done with people doing meditation so there's the idea that you know you track your thoughts that come up and then you sort of capture them but the question is well what's going on when i'm in the flow what's going on in my pre-reflective state when i'm in action and interaction a bit like now where in many ways that's what most of my life is and most people's life is and actually where maybe a lot of the interest is because that's where we struggle or have opportunities so i think this um question about how also is how are we when we're in this other sort of dynamic and how can we go back and revisit or explore what it is to be in action and what's going on there so i think that that this these same questions sort of feed back into that um related to the flow state and what it means to be a self are we most in touch with ourselves are we accessing ourselves being ourself most when we're writing a personal statement or when we're really focused on you know downhill skiing or a chess game the kinds of things that suzette mahali wrote about in flow and so during that downhill skiing or the chess game maybe somebody is not thinking about some of their somatosensory or their narrative self there's aspects of self that are actually

specifically turned down to like almost zero and then there's a different focus but then maybe somebody might self-report after the skiing i'm so happy that's when i feel like myself and so what is this self that we're getting in touch with and how is that related to our um uh spontaneously arising thoughts blue and then anyone else so just to kind of comment on that and then tie it back into um like like what who what is the self that's doing the modeling right so the flow state is super interesting and it's like that is the space where the ego kind of drops away right like when you're in downhill skiing like you're not thinking about like what my colleagues thought about what i said at the business meeting like it's totally not at all in your in your like mental it's like all you're doing is focused on that task and especially in kinesthetic um like situations like dance and drumming and and all of these things which also tend to like alleviate the trauma experience right like so all these like where the flow state happens the like that's where there's really not that ego so there's not that um like self doing the modeling but then the ego might go back and be like i'm an awesome downhill skier right like so so there's this right like self-reflection on the flow state that that is like is that the integrator right so it just kind of maybe gives more um like rise to the ego doing the narrative or the modeling when you're doing that self self modeling or or self narration right yep really interesting i'll throw question from marco in the chat marco wrote how about taking mutual inference more seriously and posit that mutualistic interactions are legitimate perceptual entities which may possess the potential to realize a markov blanket in turn what is the maximum markov blanket like i noted last session i don't think that the x or so kind of the logical boolean gate attitude to the schema and categories of realist instrumentalists um uh basically uh but i just can't read it live but yeah like it's should we be looking for a disjointness between realism and instrumentalism um that's kind of the topic of this paper how do we take a realist approach to the instrumentalists um is almost what marco is uh getting out there but anyone can take a thought on that um thomas and then in s um i was mostly going to ask where is the chat where could i perhaps read this myself uh to try and sort out what exactly is uh oh it's the youtube live chat so if you just go to our twitter look at the most recent post you'll find the link to there but i'm not going to put it in the jitsi chat here because it will make like a swoosh sound but just go to the youtube link from our twitter um yeah thank you yes i just want to uh sort of like address that by sort of um throwing a question at it or sort of a comment actually um i want to say that again in that regard i want to just point out that we the neuroscientists are the only ones who have the cake and want to eat the cake and that means amongst every scientist that is working with models across the world in every single science we neuroscientists are the only

ones that are realists very nice and i'm gonna take that to this next question so as i understand that it's almost like there's the venn diagram of people who are studying models or using models and basically no one has a problem with oh yeah the flow of water through this dam is going to be modeled with this kind of an oscillator everyone's pretty instrumental about and then you have people who are thinking the brain and there it's kind of like the stakes are high because it's it's us on the table um dissecting ourself and so it's hard to step away from a realist interpretation from what's you know right inside so how do we step outside of that and then this is where i um put with uh alex's uh suggestions and prodding about where engineering fits into this and almost is engineering like radical when we're dispensing with any kind of realism and we just want to get it done within the time frame and the budget and the affordances that we have is that akin to engineering so blue steven and then anyone else so i i love that neuroscientists have their cake and want to eat it also and i just was like thinking and reflecting like maybe there are actually some other groups where that applies like medical doctors um also like kinesthesiologists right so anybody that kind of studies like the brain or the body like does it apply like they have the model they're also using the model like maybe they're not modeling the ma or modeling the model like within the model i don't i don't know like it's like it gets like very nested with neuroscience but but there maybe are some other um groups that are doing this as well great stephen i would i would actually question that because i'm having some challenges around working with complex adaptive system applications in other fields you know like social development and stuff and i think that this is the they may be the only ones that try and get the cake and eat it but everyone is just swimming in cake mix in a funny way because everyone assumes you can make a system model and that the system model can be a snapshot and you have flows of real things flowing it's very entity based and even the idea of the self and um the the idea of the self and there's a good paper on this by bolus and where he talks about the self is actually a social construct in itself about two and a half thousand years ago the idea of self was constructed and i have a sense of so i'm in a society and if you go back to indigenous approaches i you are one of the land you and the land are one which is maybe a bit more like reality if we're gonna say reality but there is this challenge because everyone's using systems at some levels to make the models or we talk about that way so i kind of agree but i would also say that it's the the there is a paradigm challenge here about how to think about um what's going on in terms of how we are actually knowing about the world thanks steven um blue and then anyone else so that just reminded me like the self as a social construct um it's a super interesting point and i don't know if you guys

are familiar with the work of jay garfield but he talks a lot about the work of um like or the development of the self like like becoming self aware or self like in terms of like a developmental aspect and like you don't realize like like the context that you realize that you are a self like that's distinct it's actually not yourself that you realize first you actually realize others so he talks about second person so like you as soon as you realize as an infant that there's a you who are going to come take care of me like there's a you out there that are going to feed me there's the you out there that are going to make me comfortable when i'm you know sick or wet or whatever like in in very infant hood in the human species as soon as we realize that there's a you that's when the dichotomy appears that there's me right like so it's just very interesting because it it starts as a social construct this idea of the self from a very very early age thanks blue and alex v yeah thanks i want to add from engineering domain for now the system engineering have a sad joke that we can we can't include people to the scope of the system because we can't control people so okay we can control almost everything but not people in the loop so we are trying to go out of the way with it but but if engineering is a radical instrumentalism we can consider active inference framework which provides as it was said on agent level some models which provide this opportunity to include people to the scope of the system uh because and by organizing some communication structure around this agent and create some niche for them where we can communicate in uh some language of the domain we are working on and here comes to replace all stuff with languages and with ontologies and also for ontologies and knowledge life cycle management uh and it's if we consider it in engineering way and not trying to go uh down from agent level uh but trying to see or on agent environment and maybe it's as it was told that we can say about agent level how mind level and for this mind level uh the behavior of the agent is a correlate for his all kind of states under his marco blanket thanks alex it's almost like because you can't engineer the human you're engineering the inputs and the affordances that's where active inference comes into play it's like if you're going to do water engineering you're talking about making the canal and the dams and the measurement devices not going in there and engineering the water water interactions so engineering and instrumentalism is almost always at the periphery of an uncontrollable system interfacing between our regimes of order and attention and then a system whose dynamics are being modeled instrumentally because it's um the fringe it's what's not known so um ines and then dave yeah i think that's um that's about right because so we have a tendency to think that to take the realist approach if we come from the sciences of the mind and brain but then we would not commit the same kind of um would not have be have

the same kind of attitude if we were trying to model uh a non-adaptive um system so i think that comes from the fact that we do see that there's a difference between adaptive and non-adaptive and that sort of like gives us um part of the motivation to think that because they're adaptive then they must be uh using the models that we are in order to explain them but i think this is a fallacy that sort of conflates um uh the the the explanation exponential so um but the point that i wanted to make uh was more of um coming back to the self as a social construct we can push that even a little bit further if we go back to matarana and varela with the autopay as this theory and i find this very useful uh because it's it's it's a nice exercise all of us take um cognition to be something that is an ontological property of uh the organism and this is precisely what matarana specifically uh challenges with the autopias's theory it tells us that cognition is a construct that is in the eye of the scientists so cognition is not even something that has a reference um in the world which is super interesting because when we started constructing things like this that this is going to have direct implications for the way we see models uh what it tells us is that we observe behavior in uh in systems as they interact because they are adaptive systems that interact with the environment and we observe certain behavior and make sense of it by by coming up with patterns um that's how we make sense of this behavior we come up with patterns and then we deem these patterns or this behavior as cognitive but we do this as scientists the system itself the organism itself is not cognitive this is not an ontological property of of the system so i think this is this is important to think about because this also allows us to now by taking the kind of same reasoning then we can apply this by saying that we can model we can observe this this behavior we can patent this behavior through the modeling techniques that we have but the modeling technique is not an ontological property of the organism thanks for that dave and then anyone if you want a real grasp of where a self comes from you got to start with the newborn alum shore did some very detailed work on the creation of the self as part of the mother newborn dyad and the really intricate um not self modeling but us modeling and a constant dialectic of um alienation followed by reparation you know mommy dis disappoints the baby and it's the worst thing in the world the baby wants to kill mommy and then wants to eat mommy and wants to save mommy from the monsters well once you got a good grasp of that then you go and do uh ethological work and see how other creatures differentiate their children there's a lot of socialization a lot of creation of of the style of a newborn cult even though they can get up and run around pretty fast and then you come back and look at pathological cases with humans melanie klein was the first person who realized that you can psychoanalyze a child who cannot speak yet and really

worked at that and figured out how you can find out what's going on in the kid who's the who's the the baby mad at by seeing how he plays with uh toys how he plays with dolls um his pictures these kind of kids that can't uh speak can still draw and you know rip up pictures that they don't like and um there's a lot of poignance in the emergence of the self-creation of a self cool well um to kind of operationalize that and bring it back to a figure and a question in the chat this nested markov blanket uh self concept or in the neighborhood of someone writes in the chat can we measure markov blankets the way we measure or estimate any other physical entity i've heard carl fristen saying that every electron does have a markov blanket can we measure statistically speaking electrons markov blanket and i i think it might be related to some other points that marco has in the chat which we can bring up afterwards it is really important if we're talking about markov blankets um and that's figure one is this this pink dashed line that um is uh and if that's how we're going to be modeling in the free energy principle how do we do inference on these markov blankets what does have this and um how do we go from there so um shannon and s yeah so this is a good question we we we can with a caveat measure the markov blanket of of a certain electron or certain neuron it's the s statistical boundary here and it's any states that you need to describe to in order in order to explain the behavior of what you want to model and you can kind of decide as a scientist at what level do you want to describe this behavior so your electrons markov blanket might not include the behavior of the body it might just include the neighboring electrons next to it or as far basically as far out as you need to explain the behavior at the level that you want to explain it so it's not a fact of of nature which is me agreeing like wholeheartedly with this instrumentalist it's where you want to draw the boundary in order to explain the behavior that you're interested in thanks shannon and us yeah that's precisely right um i wanna just to say that uh marco blankets are not something that exists out there in the wild um so when we say that we can measure the markov blanket of a neuron it's i would probably just like formulate it differently we can apply um markov blanket as a statistical tool uh to the level of description of neuron which is the one i'm interested and it is precisely this special tool this statistical tool which does not have to correspond to physical boundaries of anything because um we get markov blankets by identifying the influences that certain states have in other states so that's how we find markov blankets they don't exist they are a way of us to um sort of like have a nice topology of how certain states are influencing other states and in the end we got would also get like a nice thing which is like how a certain system is coupled with another system and because of the it's skill-free um as uh shannon was uh rightly pointing out because of the

scale-free property of the markov blankets we can apply this statistical tool to whichever level of description we are interested in studying so we can apply this to the single-level neuron but we can also apply this to a column we can apply this to regions can apply to the brain as a whole we can apply it to the organism that is uh coupled with the environment and we can also apply this to social networks and things like that as long as there are interactions and influences um uh that are occurring between states then we can um we can apply this to explain how uh these influences are occurring um between the states and yes this would be a very instrumentalist reading and i think the only possible one cool and and marco um adds a few more points and says i don't buy the exclusionist categorization when it comes to fep because it functions both as a modeling technique and an ontological system and i think we're um hopefully conveying what you're writing and it's a discussion that is non it's a yes and with realism and instrumentalism and so um i hope we're bringing a lot of perspectives to bear on maybe where the fdp sits in this space and that's some of the great topics that were broached in the paper and some of the really important work that's being done so that we don't preemptively come hard on one side or have an exclusionist perspective and part of that is actually enacting that not just having one really convincing thought that convinces us that it is way a or way b or neither a or b but actually bringing together different perspectives on the topic and then seeing where that goes so thanks you know as always for participating which is actually enacts the pluralism through participation so stephen and then thomas we're kind of one good thing about this paper which is opening up even bigger aspects of complex adaptive systems applications so i feel like we're we're putting a lot of bigger ideas on the edge of some other fields of practice on this work that neuroscientist is doing but i think it's really valuable there's this question of non-representational approaches um in social sciences which are kind of in that ways of trying to understand things around complex complexity has been used more and one of the big areas by michael quinn patton in qualitative research is there's this new work around principles focused evaluation which moves away from more sort of betterment or sort of final product evaluation this idea of looking at broad principles for projects which are evolving but haven't got a set of rules that can just be governed and replicated it's it's it's formed you know what it is it's not completely being formed from nothing but then you've got to work with the principles and i think that may reflect some of this um discussion that's going on here with with rules and principles because i'm now thinking well rules are probably something that's more realist and principles maybe are broader than that i don't know i think there's something around this broader area of complex adaptive systems in general and what it

means to look at them and then this is one aspect of that awesome stephen thanks a lot thomas and ines yeah um so i wanted to quickly add on to what um shannon and enes were saying um in response the measurement question and in full agreement just to really answer the question can you measure a markov blanket in the same way that we can measure any other physical system that would then be a no because the markov blanket is part of our toolkit with which we measure the system um and then on the marcos commented on the exclusionary reading of uh realism and instrumentalism and uh either to pick one or the other that's a very interesting point and i guess shows that the enumeration we had of different positions um wasn't exhaustive i had not personally really considered a pluralist account and i'm not sure exactly what that means still which is i guess part of why i didn't consider it um and i think i think that will really take some some um good spelling out of what pluralism really means here because the only way i can think of it is different strands of realism because i don't feel like you can get any form of realism if you actually take instrumentalism seriously in the sense that if you're saying well this is part of our scientific toolkit with which we're describing a system realism is cut off right then and there [Applause] but of course pluralism in the sense of let's discuss different ideas and look where that leads us is a different thing and we should definitely do that because otherwise you know we're not really going anywhere yeah that was a yeah very very interesting it's like conversational pluralism or a social pluralism exploratory pluralism is a slightly different kind of claim than the um other sub questions that pluralism could impinge upon an s and then anyone else who raised their hand okay so i wanted to address the question in the chat about whether we should be realist about free energy principle um and yeah i want to just start by saying that the free energy principle is a principle so we would expect it to have like something to say about uh a tell us of things some explanation that we would expect systems to behave in such a way and this is what we discussed last week there was a distinction between the furniture principle and process theories right process theories aim to describe processes uh that would explain a certain uh psychological or cognitive or neural phenomenon or something like that the free energy principle has something to say teleologically speaking so it has something to say about how we would expect an organism to behave towards something some ultimate goal right and this is for example uh that the system does not want to die wants to avoid uncertain states uh because it doesn't want to die so he wants to select the states that are going to be much much more favorable to um to their their existence uh in a particular situation so obviously the free energy has got these sort of like much more um is charged with with giving us an explanation about uh whichever we find the

system doing the system is going to be um very likely is going to be evolving towards uh avoiding its dissipation so that's what we get from as opposed to the the process theories where we don't really get to tell us what we get is um a description of system of processes uh in systems that minimize prediction error but let's think about this and let's see what would be a realist uh take on the free energy principle um so the furniture principle is about systems or adaptive systems it can also apply as we said to all natural systems let's take the adaptive systems here because they're much more complex and that's what we are interested on um so the free energy principle is about systems that minimize free energy expectation right precisely because they want to do that that's where we find the telos um so then i want to just to draw our attention to the fact that these explanation these tell us applies to all adaptive systems and that means that it applies to bacteria um and uh there's been also a paper with pristine and calvo that um it also applies to plants and um it also applies to systems with with nervous systems right so a realist account of the furniture principle would say that uh every single adaptive system is minimizing free energy that's a realist account of the free energy principle now the question that follows is the one that relates to the paper that uh thomas and i wrote is okay so then um our representation is going to be entailed in what level the representations need nervous systems because one because this is the free energy principle applies to all adaptive systems not only to the systems that have nervous systems so if we take the realist path about the free energy principle as something that uh systems really do they they minimize they attempt to minimize this free energy um then this applies to all uh adaptive systems so where does that leave us to the representation list account very thought provoking so maybe another way to say if i'm hearing it is like if it is the case that the brain for example is minimizing free energy so that's the realist take on the free energy principle then there have to be representations in the that correspond to free energy as being some sort of quantity that's being carried in the brain whereas to do instrumentalist free energy principle one doesn't need to have any specific um stance on whether representations of any kind exist because it's being clarified that it's just like well the linear regression exists on my calculator not in the brain yeah is it is that i hope really okay stephen and then anyone else and also yeah great points happening in the chat there's a lot happening about um for example natural kinds are they disjoint or are they isomorphic with marcon blankets there's a lot of great points that people read and consider there stephen and anyone else i think one of the advantages the free energy principle has and it's less clear once you start getting into the process theories is if if through this dynamical interf interplay between external states internal states and markov blanket

states if it's working with entropy entropy can be no dimensional so you've got this advantage that it could be happening but and you could be inferring information by somehow extracting in whatever form somehow knowledge through a change in the randomness so it's by a change in the randomness that you get a signal there isn't actually a signal you may you may not have that problem with representationalism coming up even if free energy principle is actually kind of happening because it's not clear how much it's become something with dimensions so i think that the question then is once you go beyond that into the process theory um yeah some parts of that as well are kind of dimensionless there's still that flow the nestedness that the nature of things traversed in between and then there is a sense that once you create structures that you do have physical structures and you get this realist and you have to make a choice so maybe i don't know that's just my thought on that and um it's almost like is the heart of the matter how something is or how we should act because if the heart of the matter is framed as being i won't be satisfied till i get the realist explanation i won't be satisfied with what an electron is until the heart of the matter has been reached whether it is a particle whether it is a field and then if the heart of the matter were actually framed in an action-oriented way so the heart of the matter is electrons you could go way down the rabbit hole and sub-electronic stuff but the heart of the matter is how we engage with electrons the heart of the matter with a person is how we engage with them in that case it might be possible to have adequate if not excellent policy with a very partial instrumentalist reading you could say just with the models that we had in the 1980s we had instrumentalist models of the electron that allowed us to build transistors of this power that were required for this task so just taking the engineering take and so again by kind of making the heart of the issue about how to act rather than essence and identity some of these questions that are related to representationalism essentialism internalism reductionism all of that is aligned with the heart of the issue being those topics and so it's always a countervailing argument to take a relational or an interactive stance against that broader kind of philosophical gestalt versus the action-oriented turn the pragmatic turn engineering as a philosophy or as a as a way of life and then thinking about organisms as more like engineers in their niche like ecosystem engineers niche modification evolution is selecting for good engineers not the good philosopher not the bird who thinks about how to classify the seasons but the bird who reacts to the um niche in a way that leads to continued success so it's just very interesting like almost a question arising is we talked about these axes of variation and this paper in figure one or table one laid out the two by two and so we can kind of lay them out orthogonally here but there's a lot of these dimensions and some combinations two-

way combinations or three-way combinations might be um very elucidating considering well no one has thought about this combination of yes yes yes no or which of these axes are kind of tied up axiomatically that's pretty cool to think about if anyone raises their hand we can um take it a different way i'll just go through the final little slides that we had so one slide says where does design fit into all of this is design instrumental or realist and what about technology so we kind of talked about engineering but what about design and all of its meanings and then another branch we had explored a bit was we kind of talked about it earlier today as well is how can we really distill some of these uh claims and so that that could lay it out on a sort of shared table where we could come together with all of these different philosophical perspectives and experimentalist perspectives how could we reduce this to a set of claims where someone says actually um i disagree with claim two or i would like to add claim three and so that style of getting through these topics would be pretty cool so with about you know any amount of time that we want to have left what's something that somebody would like to pick up on or head off in direction of in s and then anyone else thanks yeah i can try to unpack a little bit uh claim number two so i think that here when we say that humans do inference i think that's precisely right but we can unpack this a little bit further we can say that humans are capable in uh everyday life because they are in cultured systems they're able to engage in inference uh reasoning these logics and concepts etc that we already uh mentioned right so that's what we can do but uh humans are also scientists and they also receive training um in modeling in general so to speak and once they do uh scientists uh can also do other kinds of inference which entail much more sophisticated kind of inference such as all these tools that we've been talking about like traditional bays and and and that kind of thing so i it's important to say that in everyday life we can engage in inference that inference uh does entail um representational properties because we are trying to build up a representation on the agent level about something that we puzzled about and we can also do uh engaging this kind of inference as scientists with much more sophisticated kind of tools cool um so anyone else have any thoughts on that on this slide or let's just kind of give a quick another uh coat of paint on some of these topics a lot has been raised in this paper um i'd like to return to the roadmap and just think about what this paper did and ask the authors what their next ideas just to any level of detail or however you want to convey it what are your next exciting questions in this area or how did finishing this paper prepare you for the next lines of research or investigation or collaboration that you're going to be involved in either either thomas or ness or anyone else how is it going to change their research after reading this paper actually um i'll go i'll go first

then um well hopefully i'll soon be finishing up my phd and even more hopefully starting a postdoc to actually be able to continue all of this uh properly um but i'm now working on a paper on anticipation um and in part i am trying to put the free energy principle to use and i'm having a look into how the notion of an information geometry can help us explain formalize or the sorts of statistical patterns that agents become sensitive to when they anticipate so that's one of the lines that i'm hoping to make use of the free energy principle in an instrumentalist sense cool thanks for phrasing it within the uh framing of the paper i guess ines and then we'll kind of move into the non-author collaborator range go ahead dennis oh sorry you're muted here we go you can't hear me now yeah yep okay so one exciting thing that thomas and i are working on and as a follow-up uh together with uh with a with another colleague um is precisely on how to move forward in cognition um what would um this instrument uh would be put to a good use in understanding cognition uh as taking taking models to be in a very instrumentalist kind of way so what we want to say now is uh that moving forward then models of cognition should not um attempt to uh capture or represent the properties of cognition of what cognition is but instead the very idiosyncratic patterns of activity in a particular system um so what we are challenging is the representational models of cognition that we don't think that um makes sense because for one um cognition sorry for one cognitive scientific work is not a perspective from nowhere and for two um cognition is not a fixed structure in a closed system so we take cognition in a very kind of dynamical system stochastic system kind of way so then we are trying to now develop the sort of like models that we think would be the most appropriate models to capture these aspects of cognition sounds really interesting i look forward to hearing about how you keep the general perspective while also diving into the minute particulars stephen and then anyone else yeah this is i think it's helpful to hear this thinking about modeling because trying to bring modelling into other contexts so i'd be interested like when there was when thomas there was talking about using information geometry and to think about how to model systems and i think ines is saying the same thing is it that you're not thinking about modeling the system so you can describe the realism of the brain but it's like how can you get the dynamics so it could be useful in i don't know computational psychiatry it could when you say system you're trying to see how the dynamics reveal not necessarily a realist interpretation of the brain in the bigger system but just something useful about the bigger system is it kind of like being more applied in some way to applications in in the world or is that i'll be curious on your thoughts on that yes um yeah okay so yeah so it's basically following exactly the same reasoning which is um coming

from the view that we've come uh which is that cognition is um observed in in behavior in cognitive behavior right and we try to make sense of that but cognition is not something that is uh static is something that is dynamic and uh evolves in into a certain way because systems uh adapt because systems have this particular feature so um what we think it's important to look at is uh precisely how can we explain these dynamics of this cognitive behavior instead of from having models that aim to capture the properties of cognition in a very generalized kind of way but much more specified into how can we explain this particular map this particular um cognitive activity and these can be applied in in cognitive psychology labs but it can also apply this particular way of thinking where models should capture these dynamics and this is very much the dynamical system paradigm uh where models uh uh try to capture the um obviously this entails uh um a hypothesis and a theory about what we think is the best explanation for that particular behavior that we observed and this could be like in again cognitive psychology labs in in in perceptual learning paradigms for example but it can also be as we spoke about uh last week this can also apply to um brain data analysis because that's precisely what we don't know is how the brain generates that kind of data and then we can take different positions we can look at uh mapping the structures which is a very kind of like static kind of thing to do because we want a topology nice apology of like the brain as a map which is also very useful to have but is very descriptive so basically here we are dealing with description versus explanation because once you have a model that is a hypothesis driven then you have a theory that you want to test and that's what gets us into um all the variational base kind of area and where you need those kind of tools because we are working with a lot of theories and assumptions and each model is precisely going to be um a theory about what we think that perhaps has generated the data which is what we actually have access to so that's the kind of yeah overview here's one thought um anyone else can raise their hand on actually where this philosophy to clinical applications comes into effect so let's just say that there's some debate like about molecular reductionism that's unresolved in philosophy land and then that percolates into the clinics before the philosophy is even clarified like about genetic causation so you have genetic counselors who are doing genetic medicine while there still debates what is a gene and what is causation should we think about genetic reductionism or holism that uncertainty is actually in the research paradigm in the comparability of results in the basic research is going to lead an uh inaccuracy in communication and in confidence at the clinical level in the genetic case now with as fep we're kind of talking about neuroimaging in a big way also other antaroceptive modalities are being expanded but really a lot of the

foundation of this is in neuroimaging and neuroscience and so if there is a mess philosophically then the research is going to be less coherent the communication is going to be less clear it's going to lead to clinical inaccuracy where as if it were the case that there was a clean philosophical way at least to even frame the situation perhaps that could lead to more interoperable research agendas driven by those who do and do not wish to engage in the philosophy specifically and then increase clarity of like recommendation or of communication at the clinical level with people who are multiple steps away from the paper but we could actually reference them this exact paper for example if someone said we're hypothesizing that people who have this connectivity pattern in the dynamics of the spm it's not about how your brain's connected versus someone else it's hypothesis about a dynamical system just a thought an s and then anyone else yeah i actually just wanted to read out loud uh a quote by danielle dennett that i think just really captures what you daniel uh was saying not only in the clinical setting but also in in particular is referring to to science and i think that yeah i very much agree i think it's very insightful he says um in his book that uh let me see if i oh goodness i lost it um so you can take a second to find it yes yeah anyone else can raise their hand um in the live chat people can provide any final thoughts or questions but this is just really an awesome series i appreciate the authors for joining us both sessions for all the participants who engaged with this paper and we're already seeing how this debate like is modifying the way we think about the topics so ines oh yeah i've found the quote and um well he tells us that um uh the topic of representation should be um oh no sorry it tells us there's no such thing as uh philosophy-free science there is only science whose philosophical baggage is taken on board without examination and i think that really nicely captures what you were saying daniel nice thanks a lot for sharing that shannon i think um at the risk of bringing a new topic with less than 15 minutes um to talk but this is a big question so i don't know in in neuroscience maybe there's this just pessimistic understanding that what the philosophers and neuroscientists say once it trickles down to the public it's misconstrued so we're just like constantly reteaching the general public and if a doctor is conversing with a patient they're constantly saying in this instance this is the thing that's happening in our brain that matters for this diagnosis that we have to do to help you solve problem or something um but at a like a moral and societal level in every aspect of science this debate between whether we're looking at a true natural kind or just something that we are taking a certain stance about as scientists as a society um has huge implications for policy i'm thinking of whether gender is real or not and it has huge implications for policy that like that very detrimental like it hurts if you take a certain stance

that there's a certain gene that codes for a gender when that concept itself is is a definition that might not be a complete scientific truth but some one somewhere wants to say there's a scientific truth that gene means gender means policy means and then you just have this cycle um so even though this seems like a very you maybe we could just say ah i don't agree with the philosopher or i don't agree with neuroscience and leave the debate um the fact that the debate exists permeates into the rest of society and culture and is it's important maybe to say that science science maybe even as a whole isn't realist about the world science is instrumentalist about the best way that we can interact in our world and you do have to bring in that baggage that daniel dennett was discussing and this would also include moral baggage thanks shannon and i posted this book in studying human behavior how scientists investigate aggression and sexuality by professor helen longino so taking a course with her in grad school and reading this book the way that it framed pluralism and the sciences and the different methodologies and just laid bare in some ways how these issues it's kind of like having the cake and eating it too it's like these are the issues that are so near and dear to our social reality and so it's hard to take the right stances philosophically when it's easy um lose sight of what's really happening sometimes and just how things are and like of course it's the y chromosome it's like i mean it's an allusion to genetic reductionism but it's a little bit broader or it's different than that and it's a yes and with genetic and non and it just goes so much beyond a simple reductionist claim that it's easy to get lost so how can we even use active inference and the communication channels of public engagement to actually at least within our sub domains or informational niches at first how can we bridge that gap in a new way it's an interesting question stephen and then anyone else yeah thanks for this excellent conversation i suppose this question about whether we can not just take some realist neural correlation as our way in wasn't even an option before it was like it's pretty it was basically that was the only cake on the table so people just took it and now it's like well actually we can we've got this other way of thinking you know so the rate the way we're doing it before isn't the only way it's just it was the only way so i think that's one thing and it's helping me because um i'm seeing people in this area of mental space psychology trying to prove their field by saying say peripersonal space neurons are firing and they're not necessary they're trying to use this these brain processes as evidence you know and it's like if you don't show causal you know and i mean i even see that in my own thing i think there's a tendency for me in my work with spatial approaches how can i explain this and it's like and if you can't what you left with you're just left with well not a lot sometimes so this does give another way to make um use of uh

phenomenological wisdom i think so that's what that's what i'm taking away from this so thank you yeah thanks for connecting it to your work and experience blue and thomas so i just wanted to thank enes for that quote um i thought that that was like really deep and important and like a good understanding really of um the philosophy that's like behind uh all of science might really enable um us to go forward as a society like i've just noticed with like covid and um like at least in america and stuff that's been going on in our country like people are are upset that the science is not right right like like what happens like besides like they're very mad about like the science continually updating itself which is kind of exactly what science is supposed to do so i think maybe a good understanding of the integration with um philosophy and you know the evolution of science as as a a thing a non-equilibrium steady state that's continually being updated like as a system i think that that might um you know if it's pushed into like the mainstream it might enable people to be more accepting of science when it changes thomas thanks a lot blue yeah um i think it's it's very true that at least the way we see it now if you start talking about questions like is gender real or not that it has very um big impacts on policy and how we're going to change things um and i don't know maybe this is a typical instrumentalist thing to say but i wonder if it should matter as much if it's not just we're seeing certain patterns of say injustice and we need to address these patterns of injustice and try to create policies that remove that injustice regardless of whether the categories that um we use in describing the population that is affected by the injustice latches on to something real or not i don't think that really matters that much it's it's sort of the moment when you get into the philosophical debate about gender when you're trying to um discuss policy i think you're doing policy making wrong i think that you should be looking at is going on is there injustice in this case we can say yes there is and how can we tackle that when you get into the philosophical debate you're doing something else that's also important but not for political policy making um and i think in some ways um what um i think yeah i think stephen said um about trying to legitimize certain um parts of research by relating it to brain processes like oh the thing i'm doing is only really legitimate if i can point at the brain and say like these things are doing the thing i'm um i think you see a similar thing with um the neurodiversity groups where people latch onto this scientific framework of oh well autism is all in the brain so it's real so now you have to take me seriously and i think that's in some sense it's nice to see that you know these things are being legitimized and people are taking it more seriously in another sense it's not good because we're restricting ourselves very much to a very particular idea of what counts as real only if we find it in the brain and as we've just been talking about just looking at the brain

if you're trying to understand a broader picture of what an agent in an environment is doing is um very narrow um yeah that that was it great thanks so much for sharing that i just wanted to kind of build a little bit on that um you said like whether we think x is real there's so many issues yesterday today tomorrow is x real it becomes real when it has impact on policy and what's so cool about active inference is we even use that term policy to talk about how things matter and we use the instruments of these different models the specific ones that we're talking about to talk about how things bear upon policy and so the two things or two of the things that active inference can help us do is first off it helps us separate our preference from our affordances from observations from our policy because it's really easy when talking about things that really matter climate change all these other things it gets quite blurred with what people's preferences as far as their short-term affordances long-term teleos um and then also it gives us a forum and a shared understanding and vocabulary for actually having that realism instrumentalism debate blue to what you said about this evolving state of science developing state of science um you can veer off one side and say well it's real today and what they say is real tomorrow so maybe a realist stance lends itself towards kind of in-group thinking because if you're being told if the party tells you it's real today and the party tells you something new next year you better be a realist about at least in terms of policy versus if we take a purely deconstructionist it's all instrumentalist then some people are going to say wait but if it's not real like i mean our virus is real we hope so so what is real and so that's the kind of questions that we can have a discussion that values people's perspective modes of communication modes of thinking being different roles in complex adaptive systems together so really awesome discussion we're gonna land it on time but again thanks everyone for participating if you're listening you're also welcome to participate in a future discussion or in any other activity of actin flab and find us at activeinference.org and go from there ns and thomas we look forward to your future research and having you back for any conversations you'd ever like